THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO SECONDARY SCHOOL

FORM FOUR PRE-MOCK EXAMINATION
041 BASIC MATHEMATICS

Time: 3 Hours **2025**

Instructions

1. This paper consists of sections A and section B with a total of 14 questions.
2. Attempt all questions in both sections.
3. Each question in section A carries 6 marks while each question in section B carries 10 marks.
4. Mathematical tables and non-programmable calculator may be used.
5. Write your examination number on every page of your answer sheets.

SECTION A (60 Marks)

1. (a) Simplify where necessary then identify the number of significant figures in each of the following numbers:
   i. 0.034500
   ii. 69500 kg
   iii. 40^2

   (b) Simplify then find the LCM and the GCF of the following numbers: $\sqrt{256}$, 2^5 and $(2 \times 2 \times 2 \times 3)$

2. (a) Rationalize the denominator of the following surds expression
   $\left(\frac{\sqrt{2}-\sqrt{3}}{\sqrt{3}+\sqrt{2}}\right)^2$
   (b) Solve: $\frac{\log_2 128 - \log_2 9}{\log_2 x} = \log_2 x$

3. (a) Forty-five people were invited at a certain party. 22 guests ate rice, 20 guests ate banana only, and $2y$ guests ate both banana and rice while y guests ate none of the two kinds of food. How many guests ate rice only?
   (b) Cards are numbered 1 to 30. A card is taken at random. Find the probability that it is either a multiple of 5 or multiple of 3

4. (a) The distance between two points A $(-1, 7)$ and B $(4, y)$ is 13 units. Find the value of y.
   (b) PQRS is a parallelogram, and $\overrightarrow{PQ} = \underline{a}$ *and* $\overrightarrow{QR} = \underline{b}$.

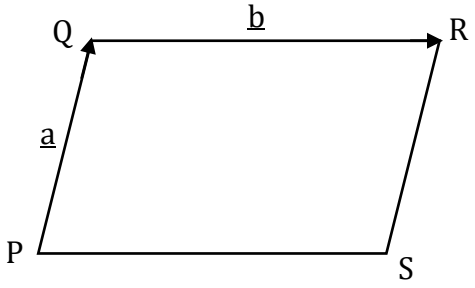


Figure 1

Find the vector expression for the following:
   i. $\overrightarrow{QP}$

ii. $\overrightarrow{SP}$

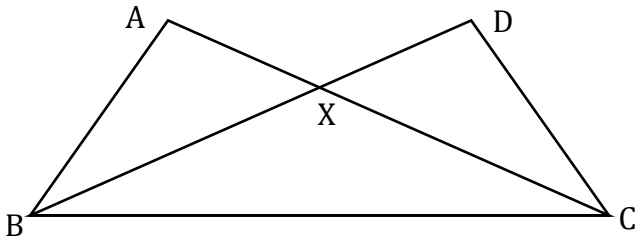
iii. $\overrightarrow{RP}$

iv. $\overrightarrow{PR}$

5. (a)A regular decagon whose sides is 3.09 cm each, is inscribed in a circle. Find the area between the circle and the decagon. (Use $\pi = 3.14$ and leave the answer to 3 significant figures)

(b) In the following figure $BAC \cong \Delta CDB$, $\angle ABC = \angle DCB \; and \; \overline{AX} = \overline{DX}, \overline{BX} = \overline{CX}$, prove that $\angle BAC = \angle CDB$

Figure 2



6. (a) Seypalm and his friend study at the same school. Seypalm lives 5.2 km from the school and his friend lives 1.8 km far from Seypalm's home. If his friend goes to school on foot every day, how long does he walk in 5 days?

(b) z varies as the square root of x and the reciprocal of y. When $x = 3, y = 4$, $and \; z = 4.5$. find the value of $z \; when x = 6, and \; y = 9$.

7. (a) Grace, Joseph, Musa, and Sarah shared 80 000 shillings in the ratio of 9: 5: 4: 2. Find 30 % of Grace's share.

(b) Use the following trial balance to prepare the income statement and the statement of financial position as at 31st December, 2019.

Trial Balance as at 31st December, 2019

Account Name	Dr	Cr
Cash	1 907 000	
Capital		2 675 000
Purchase	2 267 000	
Rent	114 000	
Furniture	305 000	
Shelves	270 000	
Sales		2 309 000
Salaries	67 000	
Wages	54 000	
	4 984 000	4 984 000

8. (a) An arithmetic progression (A.P.) consists of 50 terms, of which the third term is 12 and the last term is 106. Find the sum of the first 50 terms.

(b) Find the first term of the sequence 2, 6, 18, 54, which exceeds 10 000.

9. (a) The area of a rectangular garden PQRS is 300 m^2. If the width is 15 m, calculate the length of the diagonal of the garden

(b)Determine the value of the following without using calculator or mathematical tables

$(1 - \cos 30^{\circ})(1 - \cos 210^{\circ})$

(c) The sides of triangle PQR are given as $p = 3 \; cm, q = 5 \; cm, and \; r = 6 \; cm$. Find the cosine of an angle Q.

10. (a) Solve $3x^2 + 8x + 5 = 0$ by the method of completing the square

(b) If the side a square is increased by 4 cm, the area is increased by 40 cm. Find the side of the original square.

SECTION B (40 Marks)

11. (a) The following table shows the height of a sample of 20 Seypalm factory workers as follows:

Height in cm	$125-130$	$131-136$	$137-142$	$143-148$	$149-154$
Cumulative frequency	5	13	25	29	30

i. Find the mean height of the workers

ii. Find the mode height of the workers

iii. Find the median height of the workers

(b) Given the following figure

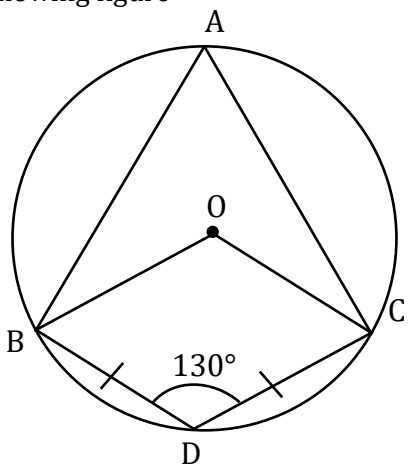


Figure 3

i. What is the name the polygon COBD?

ii. Calculate angle OCD.

12. (a) A ship sailed north from$(45^{\circ}S, 28^{\circ}W)$ to $(39^{\circ}S, 28^{\circ}W)$, taking 30 hours. What was its speed? Give the answer both in nautical miles per hour and in kilometres per hour.

(b) The following figure is a right-angled rectangular pyramid of base ABCD, vertex V, and centre of base P:

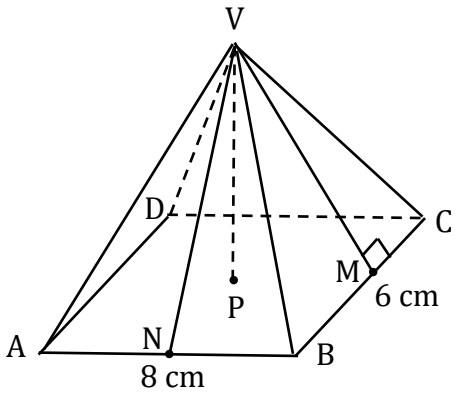


Figure 4

If $\overline{AB} = \overline{CD} = 8\,cm$, $\overline{AD} = \overline{BC} = 6\,cm$, and $\overline{VA} = \overline{VB} = \overline{VC} = \overline{VD} = 13cm$, calculate the following:

i. The height $\overline{VP}$ of the pyramid

ii. The angle between $\overline{VP}$ and the base ABCD

iii. The angle between triangle VBC and the base ABCD

13. (a) Given that $A = \begin{pmatrix} 2 & 4 \\ 3 & 6 \end{pmatrix}$ *and* $B = \begin{pmatrix} 11 & 3 \\ 4 & 1 \end{pmatrix}$, find matrix C such that $BC = A$

(b) Seypalm secondary school placed two orders with the stationery dealers. The first order was 50 boxes of marker pens and 20 boxes of correction fluid for the total amount of 40 000 shillings. The second order was 10 boxes of marker pens and 20 boxes of correction fluids for the total amount of 24 000 shillings. Calculate the cost of one box of marker pens and a box of correction fluid by using the method of inverse matrix

(c) What is the image of (1, 2) under the transformation matrix $\begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}$ followed by $\begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix}$?

14. (a) Given the graph of quadratic function. Use it to attempt the following questions:

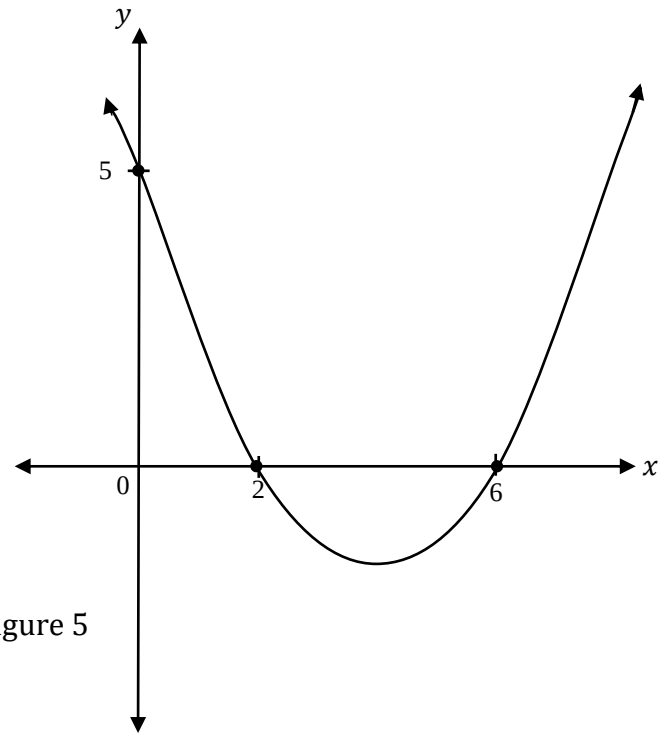


Figure 5

i. Determine the equation of symmetry
ii. State the domain and range of the function

(b) A certain local industry makes two types of bar soaps, grade A and grade B. Grade A requires 7 kg of starting materials and 1000g of finishing materials. Grade B requires 5 kg of starting materials and 3 kg of finishing materials. There are only 70 kg of starting materials and 24 kg of finishing materials available. A bar soap grade A earns a profit of 2000 shillings and grade B earns a profit of 1500 shillings. How many bar soaps of each grade should be made so as to maximize the profit?